Autocallables_research

June 22, 2026

1 Autocallables: Computing Expected Life

1.1 Quick Refresher on markov Modeling

Concept	Definition	Key criterion / formula	In the autocallable chain
State space	The set of all possible values X_n can take	—	$\{A_1, A_2, A_3, A_4, C_1, C_2, C_3, M_H, M_M,$
Transition matrix P	Row-stochastic: $P_{ij} = P(X_{n+1} = j \mid X_n = i)$	every entry ≥ 0, $\sum_j P_{ij} = 1$	each row of $[Q\ R]$ sums to 1 by construction
Transient state	Visited only finitely many times a.s.; probability of return < 1	$\sum_n P_{ii}^{(n)} < \infty$	A_1, A_2, A_3, A_4
Absorbing state	Special case of recurrent: once entered, never left	$P_{ii} = 1$	$C_1, C_2, C_3, M_H, M_M, M_L$
Recurrent state	Probability of eventual return = 1	$\sum_n P_{ii}^{(n)} = \infty$	none of the *non-absorbing* states are recurrent — that's what makes the chain absorbing
Communicating class	Maximal set of states all mutually reachable	$i \leftrightarrow j$	each A_n and each absorbing state is its own singleton class
Irreducibility	Single communicating class; every state reachable from every other		
Periodicity	gcd of return times to a state	$d(i) = \gcd\{n : P_{ii}^{(n)} > 0\}$	
Ergodicity	Irreducible + aperiodic + positive recurrent $\Rightarrow$ unique stationary π, time-avg = ensemble-avg	requires irreducibility	

Concept	Definition	Key criterion / formula	In the autocallable chain
Stationary distribution π	$\pi P = \pi,\ \sum_i \pi_i = 1$	unique only if irreducible & positive recurrent	
Canonical form (absorbing chains)	Reorder transient states first, absorbing last	$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$	exactly the Q, R split used in the autocallable formulation
Fundamental matrix N	Expected visits to transient state j starting from i	$N = (I - Q)^{-1}$, well-defined iff $Q^n \to 0$	$N = I + Q + Q^2 + Q^3$ since Q is nilpotent (chain only moves forward)
Absorption probabilities	$P(\text{absorbed in } j \mid \text{start } i)$	$B = NR$	gives $P(C_n) = \pi_n p_n$, $P(M_H), P(M_M), P(M_L)$
Expected time to absorption	Expected number of steps before hitting an absorbing state	$t = N\mathbf{1}$	expected life calculation
Chapman–Kolmogorov	Multi-step transitions compose	$P^{(m+n)} = P^{(m)}P^{(n)}$	used implicitly when propagating S_t's distribution from one observation date to the next
π	steady-state distribution	satisties $\pi = \pi$P	a probability vector where each entry corresponds to the long-term likelihood of the process being in a specific state, regardless of the starting conditions.

Reference: ORF309 lecture notes, Claude

Parameters: maturity 8 months, non-call period 2 months, observations every 2 months, autocall barrier 100%, coupon barrier 70%, maturity barrier 60%.

1.2 Setup

Treat the underlying's level relative to its initial value as the random driver. Since it is observed at discrete dates and (under standard GBM-type modeling) each date's level depends only on the *previous* date's level, the process is Markov by construction.

Observation dates: $n = 1, 2, 3, 4$ at $t_n = 2, 4, 6, 8$ months. $n = 1, 2, 3$ are call dates; $n = 4$ is maturity.

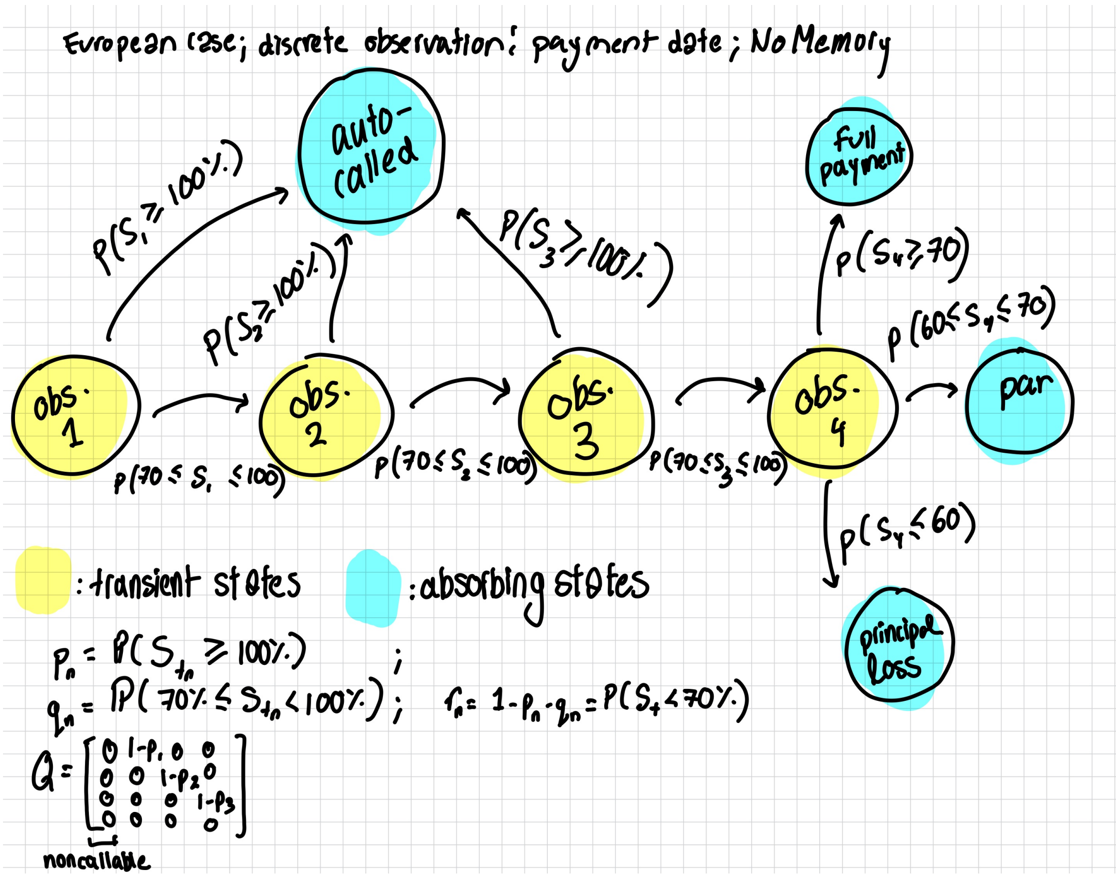
Transient states (not yet absorbed): A_1, A_2, A_3, A_4 = "alive, entering observation n."

Absorbing states: C_1, C_2, C_3 (called at observation n), and M_H, M_M, M_L (matured: full payout / par-only / capital loss).

1.3 State Diagram

[11]:
```python
from IPython.display import Image, display

display(Image("/Users/leenboualia/Downloads/ORF_335_521_copy.jpg"))
```



1.4 Canonical form and the fundamental matrix

Order transient states $[A_1, A_2, A_3, A_4]$. Since the chain only moves forward:

$$Q = \begin{pmatrix} 0 & 1-p_1 & 0 & 0 \\ 0 & 0 & 1-p_2 & 0 \\ 0 & 0 & 0 & 1-p_3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Q is nilpotent ($Q^4 = 0$), so the fundamental matrix is exact and finite:

$$N = (I - Q)^{-1} = I + Q + Q^2 + Q^3$$

N_{ij} = expected number of times the chain occupies transient state j, starting from i. Because each A_n is visited at most once, $N_{1j} \in \{0, 1\}$ along any realized path — it tracks *whether* state j is reached.

1.5 Expected life

$$E[\text{periods to absorption}] = \sum_{n=1}^{4} n \cdot \pi_n \cdot a_n$$

where $\pi_n = \prod_{k=1}^{n-1}(1 - p_k)$ is the probability of reaching observation n (with $\pi_1 = 1$), and $a_n = p_n$ for $n < 4$, $a_4 = 1$. Explicitly, in months:

$$E[\text{life}] = t_1 p_1 + t_2(1 - p_1)p_2 + t_3(1 - p_1)(1 - p_2)p_3 + t_4(1 - p_1)(1 - p_2)(1 - p_3)$$

with $t_n \in \{2, 4, 6, 8\}$. This is the expected *redemption date* — the quantity desks call "expected life," since the product rarely actually survives to the stated 8-month maturity once S has a reasonable chance of rallying.

1.6 Absorption probabilities and expected payoff

$B = NR$ gives the absorption probabilities (i.e, the probability that we end up in a specific outcome s.t $B_{ik} = P(\text{end up absorbed in state k start in transient state i})$; from A_1 these collapse to the simple cascade:

$$P(C_n) = \pi_n p_n, \qquad P(M_H) = \pi_4 q_H, \quad P(M_M) = \pi_4 q_M, \quad P(M_L) = \pi_4 q_L$$

1.6.1 Reward structure

This is a Markov chain *with rewards* — coupons attach to transitions, not just terminal states, since a coupon is paid at t_n whenever $S_{t_n} \geq 70\%$, whether or not that triggers a call.

Event	Payoff at t_n
Called at n ($n \leq 3$)	Principal + coupon c
Continues with coupon (q_n, $n \leq 3$)	c only
Continues, no coupon (r_n, $n \leq 3$)	0
M_H	Principal + coupon c
M_M	Principal only
M_L	Principal $\times S_T/S_0$

1.6.2 Undiscounted expected payoff

$$E[\text{Payoff}] = \sum_{n=1}^{3} \pi_n \Big[p_n(\text{Principal}+c)+q_n\, c\Big] + \pi_4 \Big[q_H(\text{Principal}+c)+q_M\, \text{Principal}+q_L\, \text{Principal}\cdot E\Big[\tfrac{S_T}{S_0}\,\Big|\, S_T < 60\%\Big]\Big]$$

1.6.3 Present-value (discounted) version

Multiply each n-indexed bracket by a discount factor $DF(t_n)$ before summing — this is what you would actually quote as fair value, and it is why expected life matters: it tells you *when*, on average, that PV is realized, which feeds duration/sensitivity analysis even though the PV formula itself does not need the fundamental matrix explicitly.

$$PV = \sum_{n=1}^{3} \pi_n \, DF(t_n) \Big[p_n (\text{Principal}+c) + q_n \, c \Big] + \pi_4 \, DF(t_4) \Big[q_H (\text{Principal}+c) + q_M \, \text{Principal} + q_L \, \text{Principal} \cdot E\Big[\tfrac{S_T}{S_0} \Big| \, S_T$$

1.7 Cmputing p_n, q_n

Performing this computation requires $p_n = P(S_{t_n} \geq 100\% \mid \text{no autocall at } 1, \dots, n-1)$ — a probability about the *path*, not just the marginal level at t_n. Because GBM is Markov, the clean way to compute these probabilities without approximating with unconditional marginals is to **not** collapse "alive" into a single macro-state, but instead run a discretized price-grid Markov chain: a lattice/grid over $S_{t_{n-1}}$, propagate through the lognormal transition kernel to t_n, and at each step split the probability mass at the 100% / 70% / 60% boundaries before convolving forward to the next date.

The macro-chain is the right way to *organize* the life/payoff bookkeeping (fundamental matrix, absorption probabilities, expected life); the Monte Carlo sim generates accurate numbers for p_n, q_n, q_H, q_M, q_L given a volatility, rate, and dividend assumption for the underlying.

2 Margeux and Chris's Assigment:

2.1 Autocallables

Autocallables are a structured debt whose payoffs are tied to an underlying equity or ETFs performance.

They coupon payments, early call-ability, and potential loss, are contingent on the behaviour of the underlying asset/s within thresholds pre-determined by both parties. Often, the issuer of the autocallable will sell the debt security and buy a derivative contract (put call).

Attribute	Implication
Underlying	Equity, index, basket, worst-of basket, etc that determines payoffs.
Tenor	Maximum maturity of the note if no autocall occurs.
Expected Life	Expected time until autocall or maturity. Important for hedging and pricing. $E[\text{periods to absorption}] = \sum_{n=1}^{4} n \cdot \pi_n \cdot a_n$
Observation Frequency	Monthly, quarterly, semiannual. More observation dates generally increase autocall probability.
Non-Call Period	Initial lockout period during which autocall cannot occur. Allows coupon accrual while issuer cannot redeem.
Autocall Barrier	Level (e.g. 100% of initial spot) above which the note redeems early.
Coupon Barrier	Threshold above which coupon is paid on an observation date. Usually lower than the autocall barrier.

Attribute	Implication
Protection Barrier (KI Barrier)	Level below which principal becomes exposed to underlying losses at maturity.
Coupon Rate	Periodic coupon paid if coupon condition is satisfied.
Memory Coupon	Missed coupons accumulate and are paid later if coupon barrier is breached i.e, previous missed coupons can be redeemed in the event of a rebound
No Memory Coupon	Coupons are contnigent solely on the behaviour of the underlying at the observation period, missed payments can not be redeemed.
Coupon Type	Fixed, contingent, digital, enhanced, etc.
Barrier Type	European (observed only at maturity) or American (continuous observation).
Principal Protection	Full, conditional, or none.
Leverage Factor	Multiplier applied to downside participation. Often 1x but can vary.
Worst-of Feature	Payoff determined by worst-performing underlying in a basket. Substantially increases risk.
Basket Correlation Exposure	Lower correlation increases likelihood of a worst-of name falling through barriers.
Call Premium	Redemption amount upon autocall (typically par plus coupon).
Secondary Market Liquidity	Determines ease of exiting before maturity.

2.2 Definitions

Let

- S_0 = initial underlying level
- S_T = underlying level at maturity
- B_{KI} = knock-in barrier
- N = principal

For an American barrier, define

$$M_L = \left\{ \min_{0 \le t \le T} \frac{S_t}{S_0} < B_{KI} \right\}$$

where M_L denotes the event that the barrier has been breached at least once during the life of the note.

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3 European Knock-In Barrier

The barrier is observed **only at maturity**.

Maturity Scenario	Condition	Principal Payoff
Barrier not breached	$S_T \ge B_{KI}$	N

Maturity Scenario	Condition	Principal Payoff
Barrier breached	$S_T < B_{KI}$	$N\frac{S_T}{S_0}$

Equivalent payoff representation:

$$\text{Payoff} = \begin{cases} N, & S_T \geq B_{KI}, \\ N\frac{S_T}{S_0}, & S_T < B_{KI}. \end{cases}$$

3.0.1 Interpretation

The underlying may trade below the barrier throughout the life of the note. Only the final observation at maturity matters.

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4 American Knock-In Barrier

The barrier is monitored continuously (or daily, depending on the product).

Maturity Scenario	Condition	Principal Payoff
Barrier never breached	$M_L^c, S_t >$ Coupon Barrier	N
Barrier breached, underlying finishes below initial level	M_L and $S_t < S_0$	$N\frac{\min_{0\leq t\leq T} S_t}{S_0}$

Equivalent payoff representation:

$$\text{Payoff} = \begin{cases} N, & \min_{0\leq t\leq T} S_t \geq B_{KI}, \\ N, & \min_{0\leq t\leq T} S_t < B_{KI} \text{ and } S_T \geq S_0, \\ N\frac{S_T}{S_0}, & \min_{0\leq t\leq T} S_t < B_{KI} \text{ and } S_T < S_0. \end{cases}$$

4.0.1 Interpretation

Once the barrier is breached, the downside participation feature is activated. The final payoff then depends on where the underlying finishes relative to its initial level.

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5 Comparison

Feature	European KI	American KI
Barrier observation	Maturity only	Continuous (or daily)
Relevant quantity	S_T	$\min_{0 \le t \le T} S_t$ and S_T
Can recover after falling below barrier?	Yes	No
Investor protection	Higher	Lower
Coupon offered	Lower	Higher

6 Twin-Win Note

Consists of a knock-in put, a call option, and a put option

6.1 Definitions

Let

- S_0 = initial underlying level
- S_T = underlying level at maturity
- B_{KI} = knock-in barrier
- N = principal

Define the final return

$$R = \frac{S_T - S_0}{S_0}.$$

Define the barrier event

$$M_L = \left\{ \min_{0 \le t \le T} \frac{S_t}{S_0} < B_{KI} \right\}.$$

The key feature of a Twin-Win note is that downside performance (within established knock-in levels) is converted into positive performance, provided the knock-in barrier is never breached.

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7 Payoff Structure

Scenario	Condition	Payoff
Underlying rises	$R \ge 0$	$N(1 + R)$
Underlying falls but barrier not breached	$R < 0$ and M_L^c	$N(1 + \lvert R \rvert)$
Barrier breached	M_L	$N(1 + R)$

Equivalently,

$$
\text{Payoff} = \begin{cases} N(1+R), & R \geq 0, \\ N(1+|R|), & R < 0 \text{ and } M_L^c, \\ N(1+R), & M_L. \end{cases}
$$

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8 Comparison

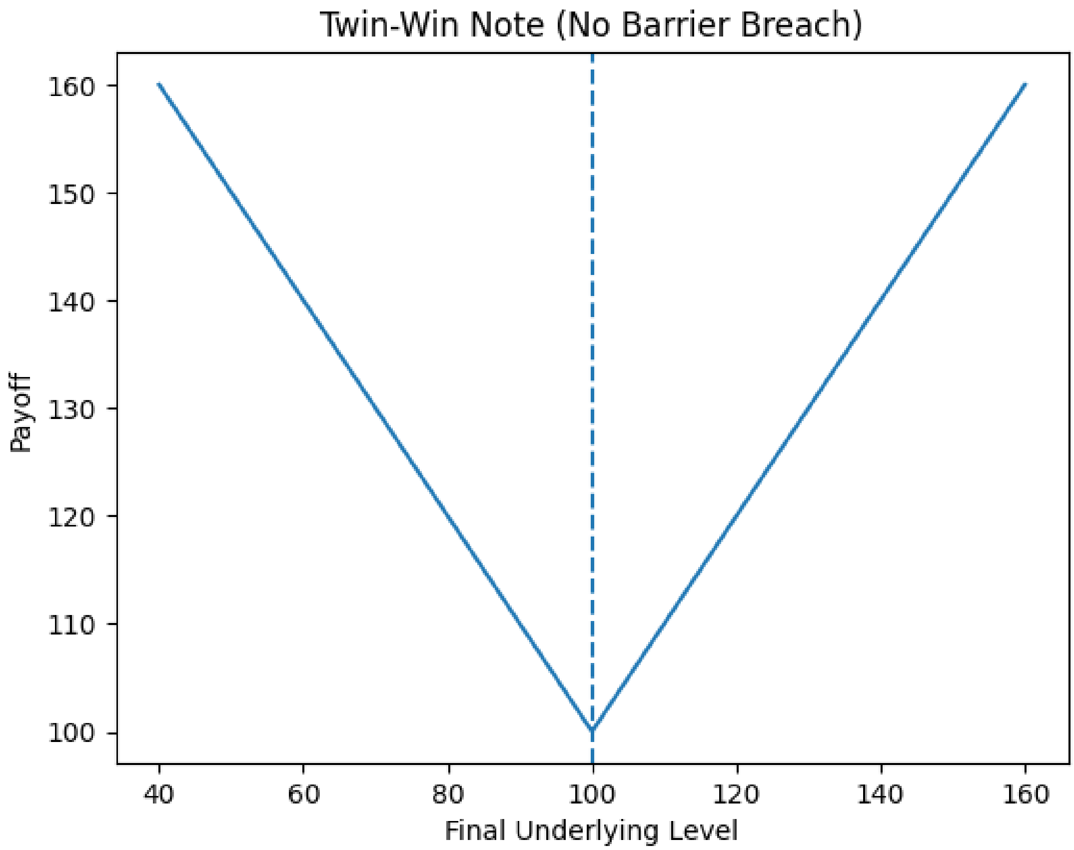
Feature	Twin-Win	Autocallable
Expected life uncertainty	No	Yes
Main risk parameter	Knock-in barrier	Coupon barrier + KI barrier + autocall barrier
Upside participation	Usually capped or linear	Usually limited to coupons
Benefits from downside moves	Yes	No

[3]:
```python
import numpy as np
import matplotlib.pyplot as plt

S = np.linspace(40, 160, 500)

payoff = np.where(
    S >= 100,
    S,
    200 - S
)

plt.plot(S, payoff)
plt.axvline(100, linestyle='--')
plt.xlabel("Final Underlying Level")
plt.ylabel("Payoff")
plt.title("Twin-Win Note (No Barrier Breach)")
plt.show()
```



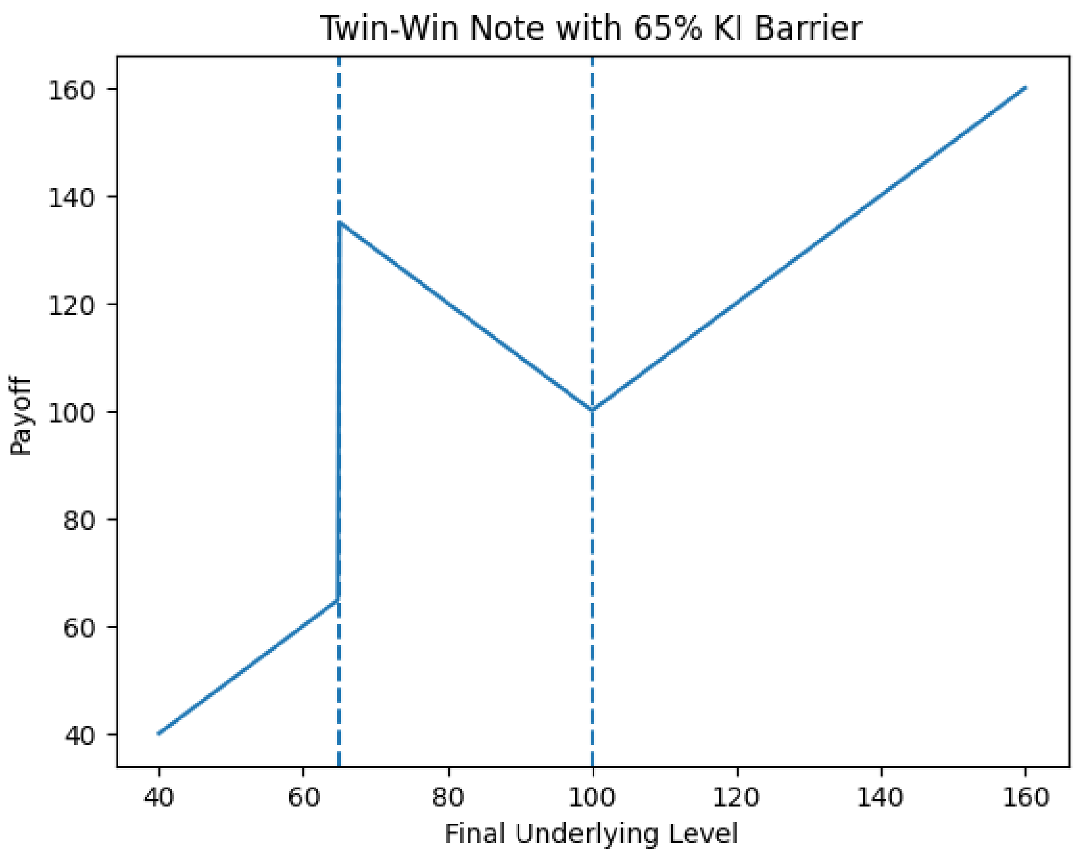
[4]:
```
import numpy as np
import matplotlib.pyplot as plt

S = np.linspace(40, 160, 500)

payoff = np.where(
    S >= 100,
    S,
    np.where(
        S >= 65,
        200 - S,
        S
    )
)

plt.plot(S, payoff)
plt.axvline(100, linestyle='--')
plt.axvline(65, linestyle='--')
plt.xlabel("Final Underlying Level")
```

```
plt.ylabel("Payoff")
plt.title("Twin-Win Note with 65% KI Barrier")
plt.show()
```



9 Airbag/Buffer Note

9.0.1 Background:

A bull call spread consists of one long call with a lower strike price and one short call with a higher strike price. (source: Fidelity). A ratio call spread is when you buy a specific number of calls and sell a greater number of calls with a higher strike, both for the same expiration

An Airbag/Buffer consists of an ATM long call and ratio call spread (long zero-strike call + short call at the airbag level)

Let

- S_0 = initial underlying level
- S_T = underlying level at maturity
- N = principal
- B = buffer level

Define the underlying return

$$R = \frac{S_T - S_0}{S_0}.$$

Suppose the note has a 20% buffer.

Then

$$B = 20\%.$$

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10 Buffer Note

The investor absorbs losses only after the underlying declines by more than the buffer amount.

10.1 Payoff Structure

Maturity Scenario	Condition	Payoff
Underlying rises	$R \geq 0$	$N(1+R)$
Loss within buffer	$-B \leq R < 0$	N
Loss exceeds buffer	$R < -B$	$N(1+R+B)$

Equivalent payoff:

$$\text{Payoff} = \begin{cases} N(1+R), & R \geq 0, \\ N, & -B \leq R < 0, \\ N(1+R+B), & R < -B. \end{cases}$$

10.2 Example: 20% Buffer

Underlying Return	Investor Return
+30%	+30%
+10%	+10%
-10%	0%
-20%	0%
-30%	-10%
-40%	-20%

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11 Airbag Note

An Airbag Note provides leveraged downside protection after the barrier is breached.

Rather than losing dollar-for-dollar beyond the buffer, losses are reduced by a participation factor.

Let

$$L = \frac{|R| - B}{1 - B}$$

denote the adjusted loss beyond the barrier.

11.1 Payoff Structure

Maturity Scenario	Condition	Payoff
Underlying rises	$R \geq 0$	$N(1+R)$
Loss within buffer	$-B \leq R < 0$	N
Loss exceeds buffer	$R < -B$	$N(1-L)$

A common equivalent expression is

$$\text{Payoff} = \begin{cases} N(1+R), & R \geq 0, \\ N, & -B \leq R < 0, \\ N\left(1 - \frac{|R|-B}{1-B}\right), & R < -B. \end{cases}$$

11.2 Example: 20% Airbag

Underlying Return	Investor Return
+30%	+30%
+10%	+10%
-10%	0%
-20%	0%
-30%	-12.5%
-40%	-25.0%
-60%	-50.0%

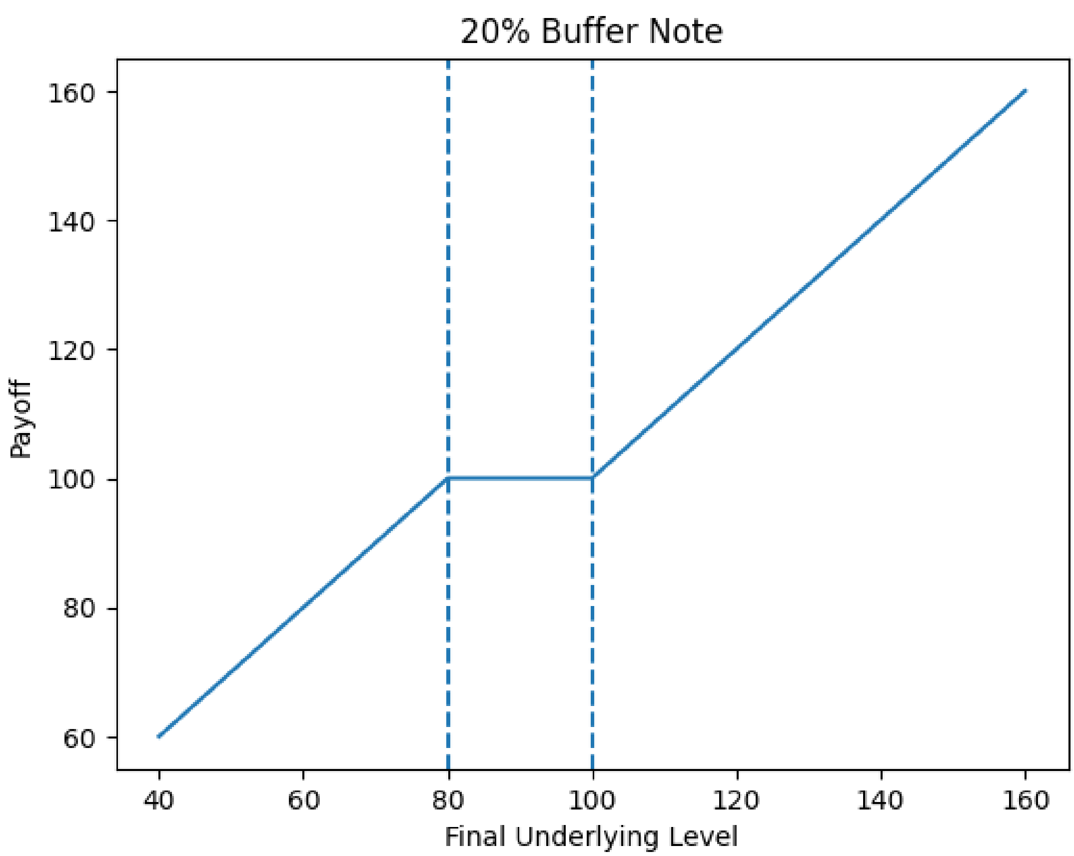
---

[6]:
```python
import numpy as np
import matplotlib.pyplot as plt

S = np.linspace(40, 160, 500)
```

```
payoff = np.where(
    S >= 100,
    S,
    np.where(
        S >= 80,
        100,
        S + 20
    )
)

plt.plot(S, payoff)
plt.axvline(80, linestyle='--')
plt.axvline(100, linestyle='--')
plt.xlabel("Final Underlying Level")
plt.ylabel("Payoff")
plt.title("20% Buffer Note")
plt.show()
```

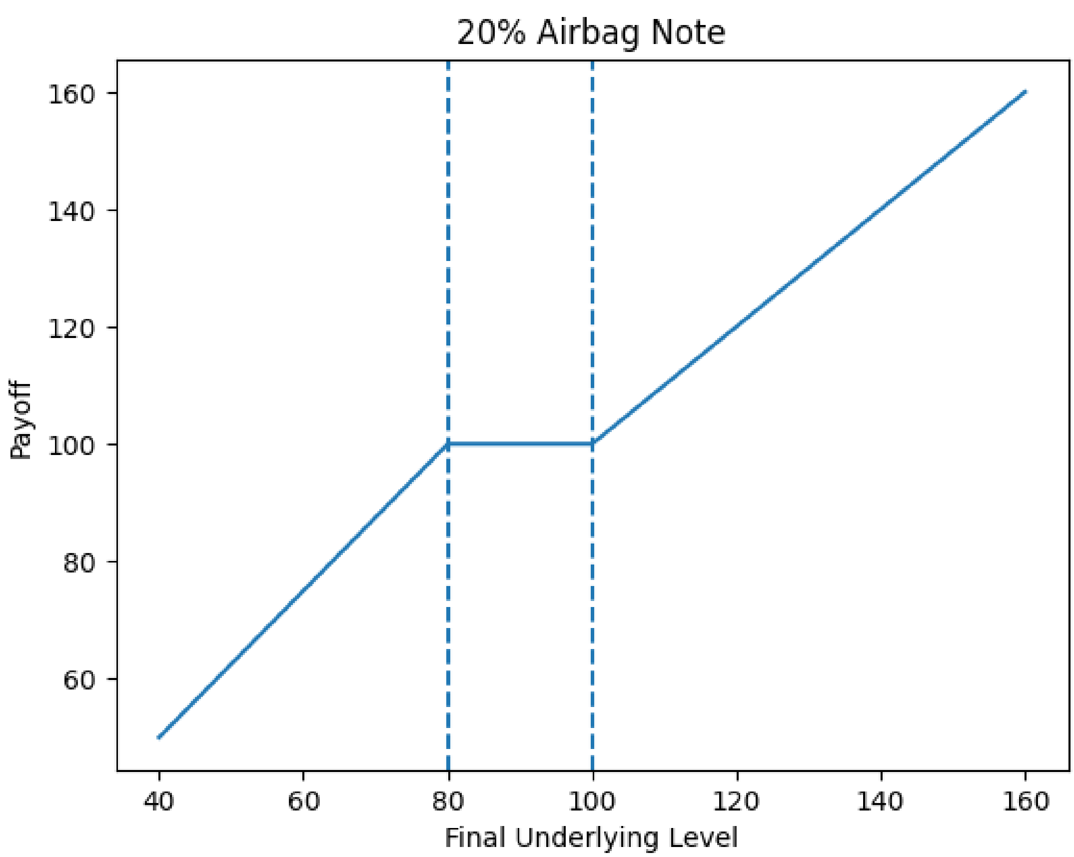


[8]:
```python
import numpy as np
import matplotlib.pyplot as plt

S = np.linspace(40, 160, 500)

payoff = np.where(
    S >= 100,
    S,
    np.where(
        S >= 80,
        100,
        100 * (1 - ((100 - S) - 20) / 80)
    )
)

plt.plot(S, payoff)
plt.axvline(80, linestyle='--')
plt.axvline(100, linestyle='--')
plt.xlabel("Final Underlying Level")
plt.ylabel("Payoff")
plt.title("20% Airbag Note")
plt.show()
```



Structured product designers can tweak airbags with: - Caps: Selling an extra call to limit upside but lower the airbag further - Worst-of Features: Linking to two assets to monetize correlation - Adjusted Strikes: Raising the strike of the call to fine-tune protection

[ ]:

12 Snowball

13 Hybrid

14 Catapult

15 Comparison

Feature	Buffer Note	Airbag Note
Upside participation	Usually 1x	Usually 1x
Downside protection	Fixed buffer	Leveraged protection
Barrier level	Yes	Yes
Loss beyond barrier	Dollar-for-dollar	Reduced participation
Complexity	Lower	Higher
Typical investor appeal	Moderate downside protection	Stronger downside protection

[ ]: